

Warmup/Attendance: [Week 5: Wednesday 9/23/2026]

Full Name:

SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. If I roll a fair **six**-sided die, what is the chance that I will roll a multiple of 3 OR a multiple of 4?

2. We have two events A and B in an outcome space Ω . If $P(A) = 0.7$ and $P(B) = 0.5$, can A and B be mutually exclusive? Draw a Venn diagram and use it to say what is the *biggest* the intersection of A and B can be and also, what is the *smallest* this value can be. That is, find upper and lower bounds on the value of $P(A \cap B)$.

3. Do you share a birthday with someone in the room? We have roughly 80 people in the classroom. Do you think the chance that you share a birthday with someone is more than 50% (just the day, for example, February 22)? How many people would you need in the room before the chance that two people have the same birthday is more than 50%? How would you even begin to think about this chance?